Akshit Tyagi

Founding Member of Technical Staff @ Axiom — ML for life sciences: multimodal LLM post-training, evaluation design, and production tooling for drug discovery

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Work Experience

Jan 2025 - Present — **Founding Member of Technical Staff** at Axiom
Multimodal LLMs and evaluation infrastructure for drug safety

- Fine-tuned multimodal LLMs over cell imaging, ADME properties and human exposure data for mechanism-of-action classification and dose-response prediction — teaching a model to read scientific images and reason about compound behaviour
- Designed the evaluation criteria the models are held to: phenotype-level and clinical-endpoint benchmarks against FDA DILIrank and against 2D HepG2, 2D PHH and 3D spheroid *in vitro* assays, so model gains are measured against the experiments they aim to replace
- Shipped tooling biologists use daily — prediction and triage over >100,000 molecules that decides which compounds enter the wet lab; co-designed the underlying multiplexed cytotoxicity assay with biologists and external labs / CROs, turning their feedback into what the models optimise for
- Built in silico clinical risk models predicting drug-induced liver injury (DILI) from molecular structure (SMILES) — 0.92 AUC (77% sensitivity, 90% specificity) on FDA DILIrank
- Own the production stack end to end: distributed Python pipelines over a 10M+ image corpus (Ray, Spark, Modal), automated batch ingestion, and drift monitoring on incoming data and retrained models

July 2023 - 2024 — **Research Engineer** at Helm AI — trained and fine-tuned UNets, latent diffusion models and DINO-family models for perception and generative simulation in autonomous driving; built semi-supervised "deep teaching" methods for foundation-model training and optimised the production SDK for on-device latency

May 2022 - July 2023 — **Research Engineer** at Google DeepMind
Google Health — clinical ML in research collaboration with Stanford hospitals

- Fine-tuned T5 transformers over longitudinal, multimodal EHR data to infer patient medical condition and forward risk from irregularly sampled clinical records
- Ran the models in shadow deployment at Stanford against live clinical data with clinicians in the loop
- Showed that aggregate metrics hide model failure: risk models with near-identical AUC disagree substantially on individual patients — an evaluation-design result, published at NeurIPS 22 (ML4H)

Nov 2021 - Mar 2022 — **ML Engineer** at Akasa — multi-modal, context-aware representations of care-provider/patient interactions from insurance cards and prior medical history

Mar - Oct 2021 — **ML Engineer** at Armorblox — time-series anomaly detection and out-of-distribution models over user email profiles for account-compromise detection; weak supervision to cut maintenance under drift

July 2020 - Jan 2021 — **AI Resident** at X, the moonshot factory — ultra-low-power TinyML for iyo

Publications and Posters

A calibration benchmark for pathology foundation models. *ML4H 2026*

Instability in clinical risk stratification models using deep learning. *NeurIPS 22 (ML4H)*
Daniel Lopez-Martinez, Alex Yakubovich, Martin Seneviratne, Adam D Lelkes, Akshit Tyagi, Jonas Kemp, Ethan Steinberg, Lance Downing, Ron Li, Keith Morse, Nigam Shah, Ming-Jun Chen.

Fast Intent Classification for Spoken Language Understanding. *ICASSP 2020*
Akshit Tyagi, Varun Sharma, Rahul Gupta, Lynn Samson, Nan Zhuang, Zihang Wang, Bill Campbell.

Conversational Question Answering and Noise Robustness. *Amazon Research Days 2019*
Akshit Tyagi, Anjishnu Kumar, Abhinav Sethy.

Selected Research & Projects

Independent	**Context Harness for LLM Agents** Built a domain-agnostic harness around LLM calls: it assembles context under a hard token budget, resolves stale and contradictory records before the prompt is built, routes across model tiers, lets the model recall dropped context through a tool loop, checks each answer deterministically and traces every decision. Two domains run on the one core. A lab research assistant for a scientist purifying a protein handles plan drift, notebook entries that conflict with instrument readings, and a sensitive record whose constraint always ships. A personal-health assistant over Epic records (FHIR R4, C-CDA) and wearables enforces in code and tests that EHR content never reaches a model API.
Independent	**A Calibration Benchmark for Pathology Foundation Models** — ML4H 2026 Foundation models are increasingly used as frozen feature extractors under exactly the conditions — scanner, stain and site shift — where confidence decouples from correctness, and a model that cannot signal its own uncertainty is unsafe to build a scientific workflow on. Built the benchmark end to end: task suite, shift conditions, confidence–correctness metrics, and an analysis of how much post-hoc recalibration recovers and where it fails.
Axiom	**Structure-to-Risk Modelling and Assay Co-Design for DILI** — presented at SoT 2025 Profiled >100,000 molecules in primary human hepatocytes under a multiplexed assay combining cell-painting imaging with LDH and MT-Glo readouts. Trained models predicting dose-dependent responses across >10 cell stress and death phenotypes from SMILES alone, then fine-tuned a multimodal LLM over cell images with ADME and exposure data to yield a clinical risk score benchmarked against 2D HepG2, 2D PHH and 3D spheroid assays. Co-designed the assay itself with biologists, feeding model output back to identify which phenotypes carry biological signal — a working loop between what the model can learn and what the experiment measures.
Axiom	**Cellular Imaging Infrastructure** Production infrastructure behind the modelling work: microscope-level correction so instrument variation doesn't propagate downstream; automated ingestion and normalisation of each batch into a growing 10M+ image corpus; drift monitoring on both incoming data and retrained models; segmentation quantifying vacuoles, cytoplasm, mitochondria and RNA condensates across cell-painting channels; and fine-tuned DINO embeddings clustering phenotypes to surface bioactive molecules.
Google	**Clinical Risk Models with Stanford Hospitals** — NeurIPS 22 (ML4H) Built T5-based models over longitudinal EHR data with clinicians at Stanford and validated them in shadow deployment against live clinical data. Showed that risk-stratification models with near-identical aggregate performance can disagree substantially on individual patients — an instability that has since grown into its own line of research.

Technical Skills

LLMs	Multimodal LLM fine-tuning & post-training (LoRA, RLHF, DPO, SFT data curation); vision foundation models (DINO); diffusion models; segmentation; LLM harnesses (context assembly, memory, tool use, tracing)
Evaluation	Benchmark design; calibration & uncertainty quantification; robustness under distribution shift; eval-driven model development
Biology	Cheminformatics (SMILES / InChI); ADME and exposure modelling; high-content microscopy and cell painting; longitudinal EHR; assay design with biologists and CROs
Engineering	Production Python; distributed pipelines at 10M+ image scale (Ray, Spark, Modal); Docker, GCP, AWS; PyTorch, Hugging Face; Weights & Biases, MLflow; C++, CUDA

Education & Service

M.S. Computer Science, UMass Amherst (2020) · B.Tech. Electrical Engineering, IIT Delhi (2018) · Awards: DISA (IIT Delhi), Institute Award (top 7%), KVPY Fellowship · Reviewer: NeurIPS, ICML, ICLR, ACL